

Armature reaction and commutation process in DC machine

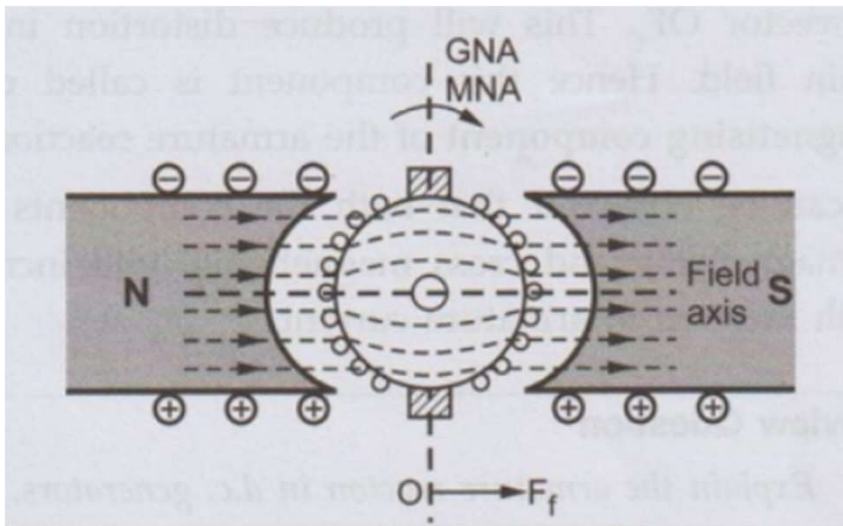
MODULE 1

Electrical Technology

Armature reaction

- The effect of magnetic field setup by the armature current on the distribution of main pole flux of a DC machine is known as armature reaction
- The armature reaction produces the following two effects:
 1. It demagnetizes or weakens the main flux.
 2. It cross-magnetizes or distorts the main flux.

Polar Axis, GNA, MNA



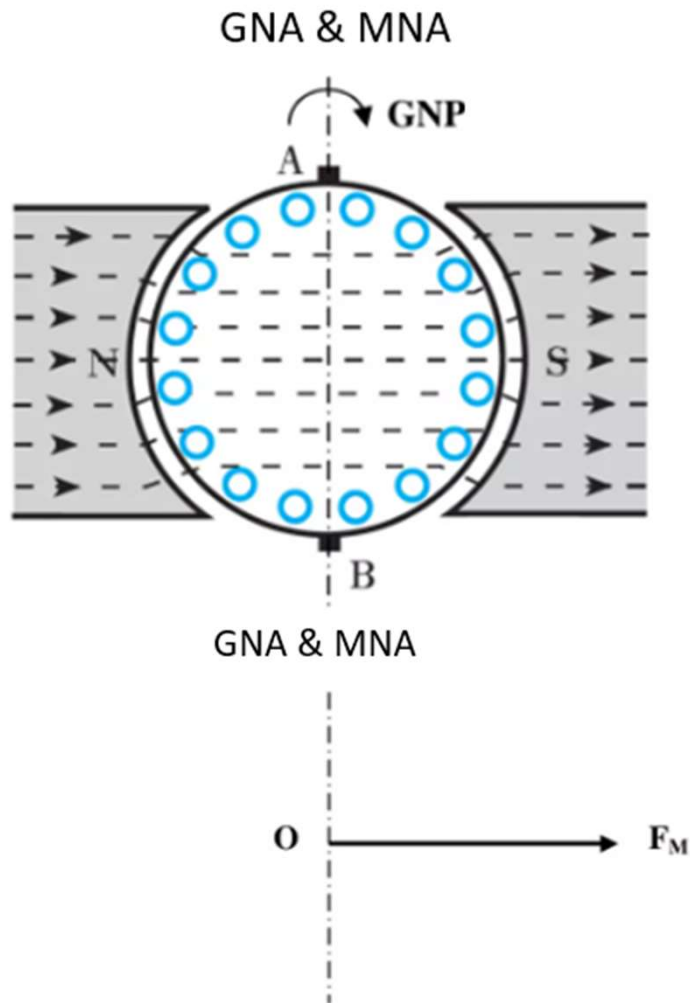
- Polar axis is line joining the centres of N and S poles.

- GNA – Geometric neutral axis
- The axis perpendicular to the polar axis is called GNA
- GNA change with load.
- GNA is the axis of symmetry between the adjacent poles

MNA – Magnetic neutral axis

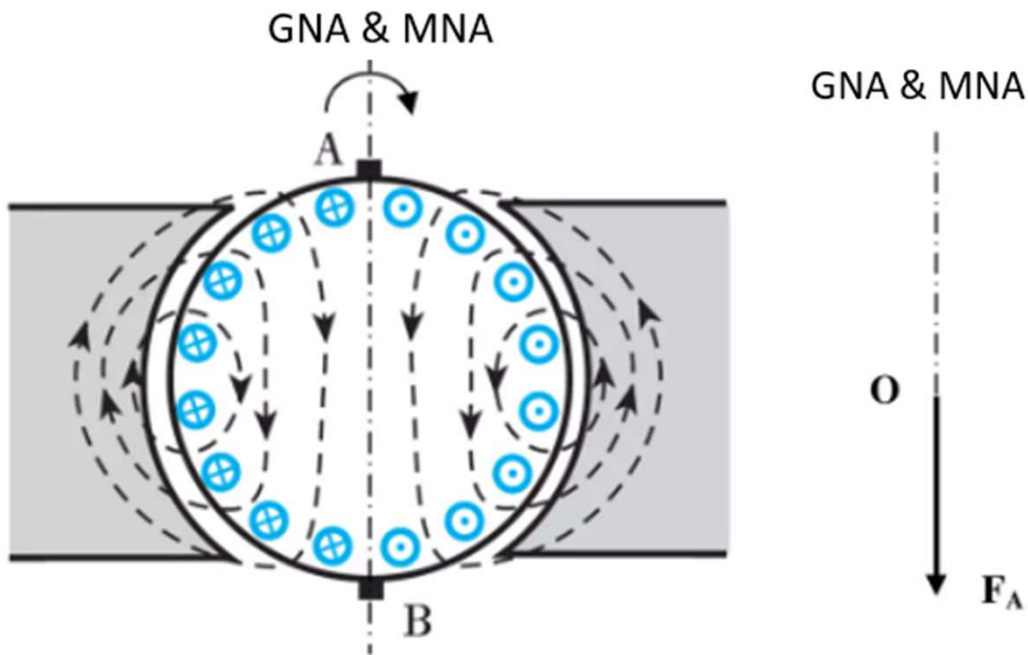
- The axis along which there is no e.m.f. induced in the armature conductors is called. Magnetic Neutral Axis (MNA).
- MNA can also be called 'axis of commutation' since reversal of current takes place along this axis.
- MNA is always perpendicular to the main field

Distribution of flux by main field poles (with machine on no load condition)



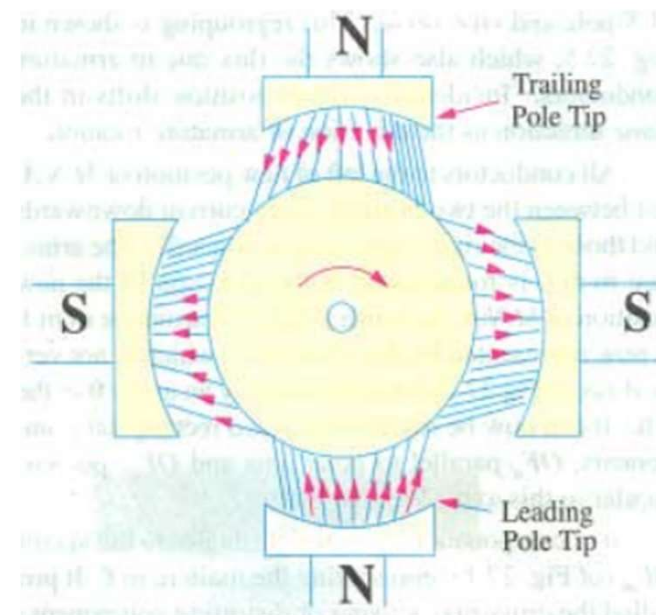
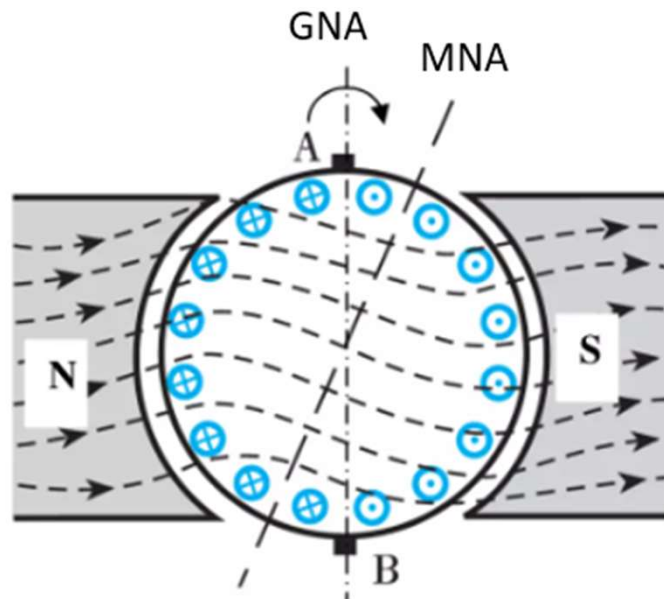
- The flux is distributed symmetrically with respect to axis called polar axis. The direction of flux produced by field coil is from left to right through an armature core and lines of force through the core are parallel to field axis.
- Magnetic neutral axis (MNA) and Geometric Neutral Axis (GNA) coincides with each other
- Brushes will make contact with the conductors lying in MNP
- MNP is the plane through the axis along which no emf is induced in the armature conductors

Distribution of flux due to armature current carrying conductor while field coils carrying no current

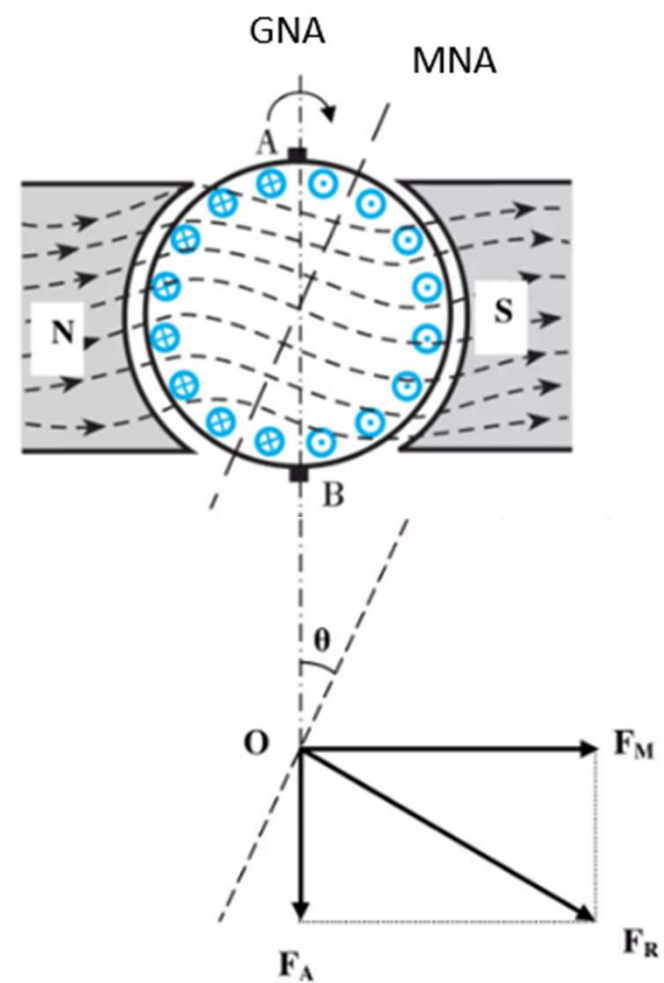
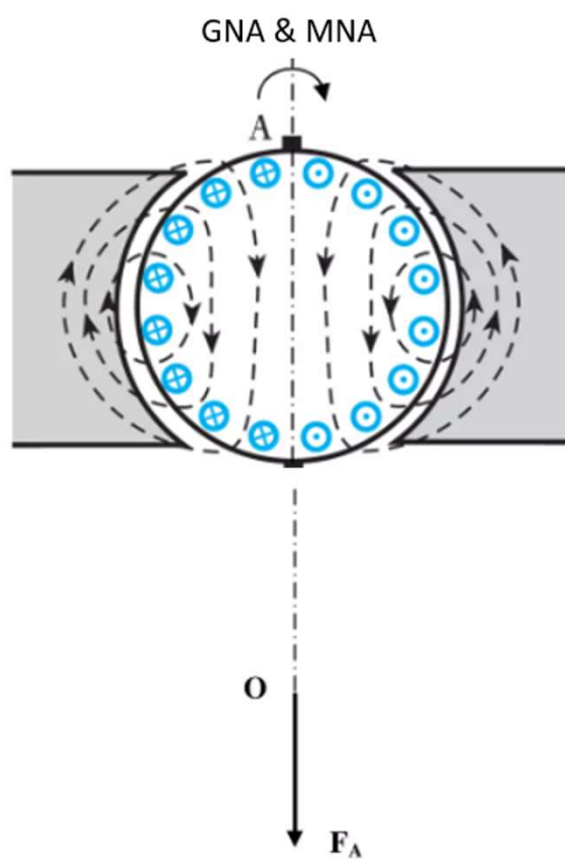
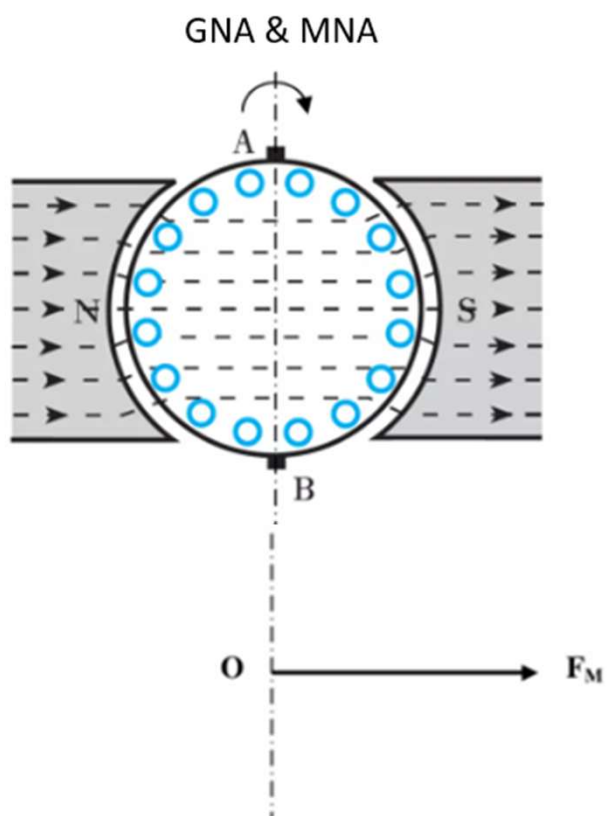


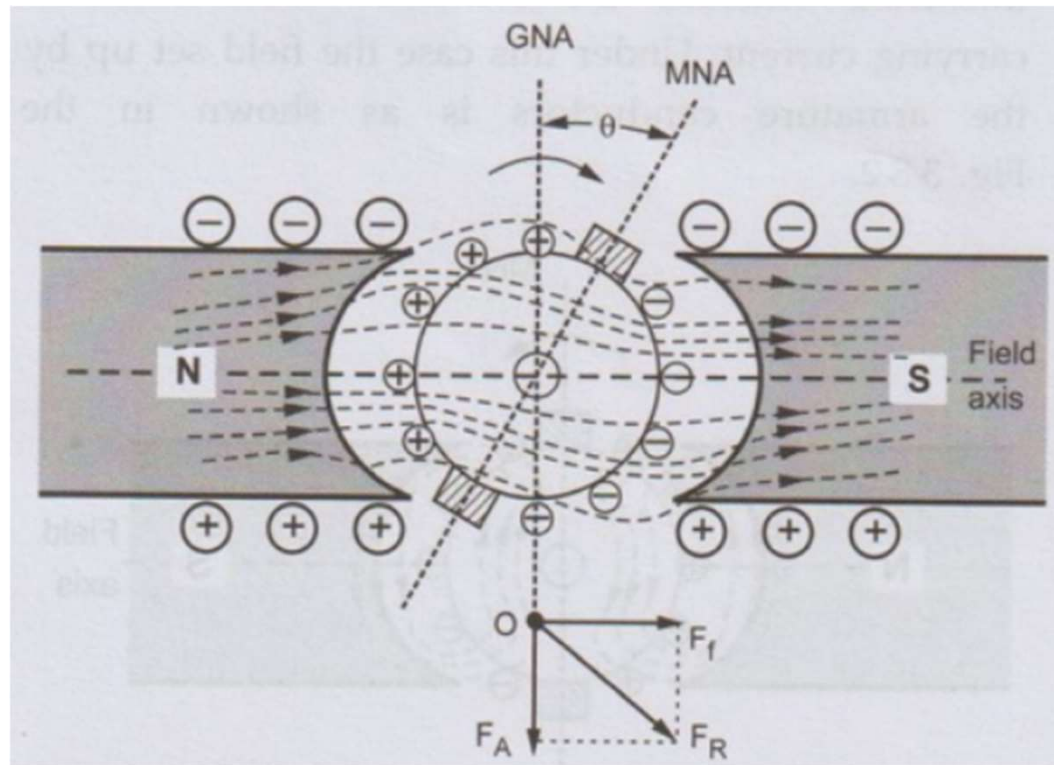
- The direction of the flux produced by current carrying conductors is vertically downwards in the armature core.
- This flux is symmetrical about a brush axis. In other words current carrying armature conductors try to magnetise the armature core along the brush axis.
- The vector OF_A represents armature m.m.f. both in magnitude and direction. This m.m.f. depends on the magnitude of the armature current.

Distribution of resultant flux established due to the simultaneous action of field and armature current



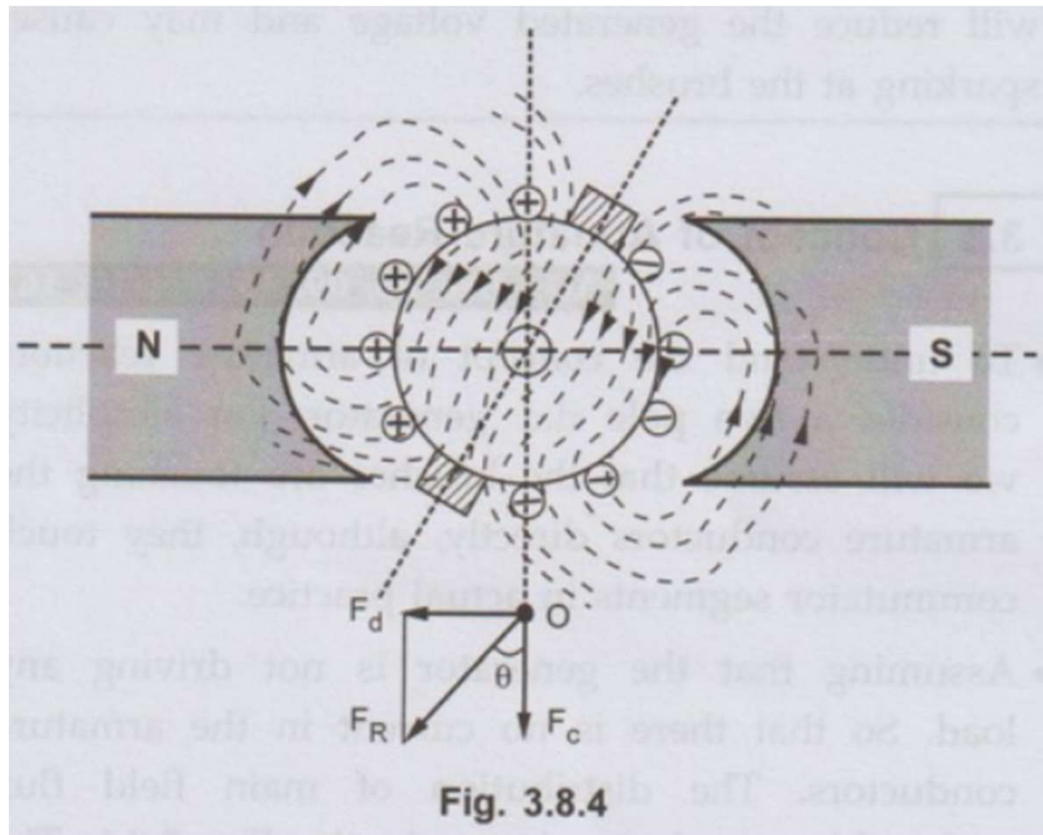
- Each line of forces produces by armature crosses the air gap twice and hence it is observed that the flux is crowded or concentrated at the trailing pole tips but weakened out at the leading pole tips
- The magnitude of shift depends on the magnitude of armature current





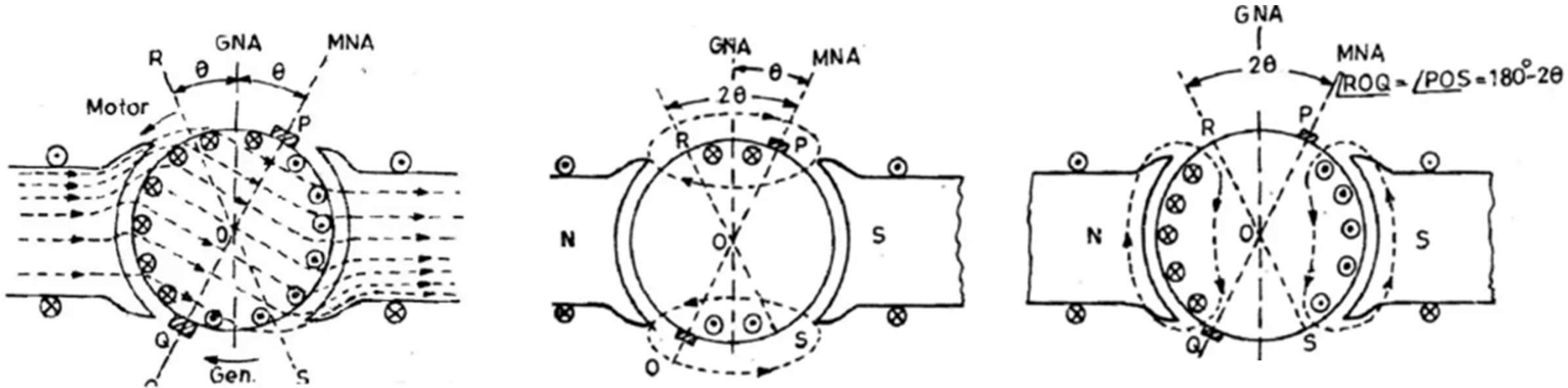
- The resultant m.m.f. OFR can be found by vectorially combining OFm, and OFA
- The new position of MNA is also shown which is perpendicular to the resultant m.m.f. vector OFR. The MNA gets shifted through an angle θ so that brushes are also shifted and are along new MNA.
- Due to this brush shift, the armature conductors as well as armature current is redistributed. Some of D.C. Generators the armature conductors which were earlier under the influence of S pole now come under N pole and vice versa.
- The conductors on the left of new position of MNA carry current downwards and those to the right carry current upwards. The armature m.m.f. is now along new position of MNA represented by vector OFR. It is inclined at an angle θ to the left instead of being vertical.

Demagnetising and Cross magnetizing effect



- The armature m.m.f. represented by vector OF_R can be resolved into two components, OF_d parallel to the polar axis and OF_c perpendicular to the axis.
- The component OF_d is in direct opposition with field m.m.f. vector OF_m . This will tend to reduce the total flux. Hence this component is called demagnetising component of the armature reaction.
- OF_c is at right angles to vector OF_m . This will produce distortion in the main field. Hence this component is called cross magnetising component of the armature reaction.

Demagnetising and cross magnetising conductors



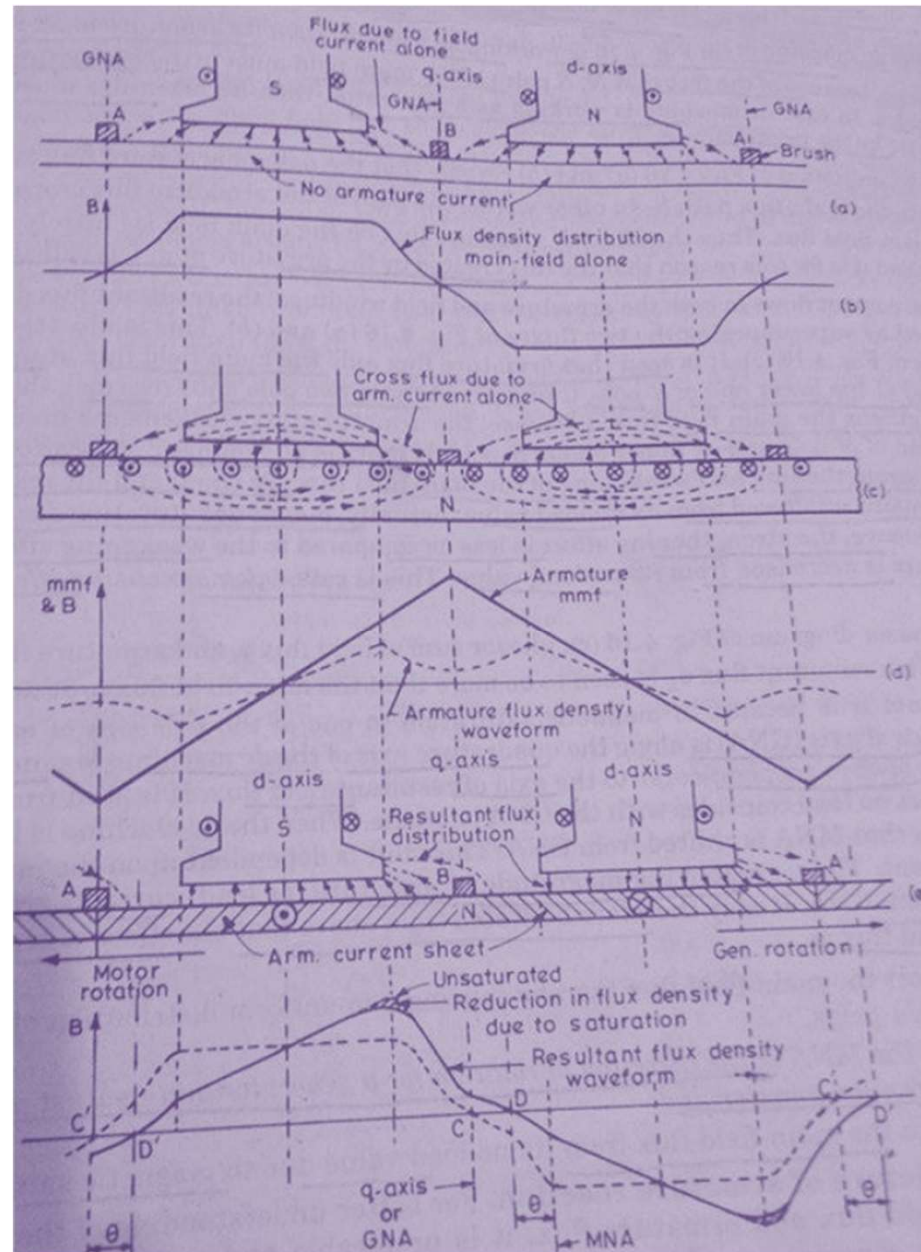
- Demagnetising ampere turns/pole $AT_d = \frac{Z\theta}{360} I_c$
 - Demagnetising ampere turns/pole $AT_d = Z I_c \left[\frac{1}{2P} - \frac{\theta}{360} \right]$
- P = no. of poles, Z = total number of armature conductors,
 I_c = Current per conductor = I_a/A

Cross magnetising effect

- Proportional to magnitude of armature MMF
 - Shift the neutral plane along the direction of rotation
 - Creation of magnetic field in the interpolar region where brushes are placed for commutation
 - Weakened field strength in air gap under leading pole tip and strengthened field under trailing pole tip
- I. Increases Iron losses at load
 - II. Poor commutation

Demagnetising effect

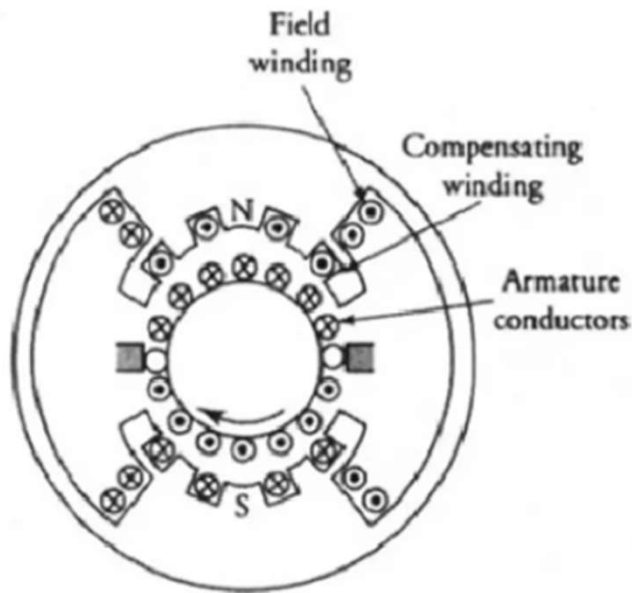
- Reduces the total flux per pole from its no load value



Remedies for field distortion

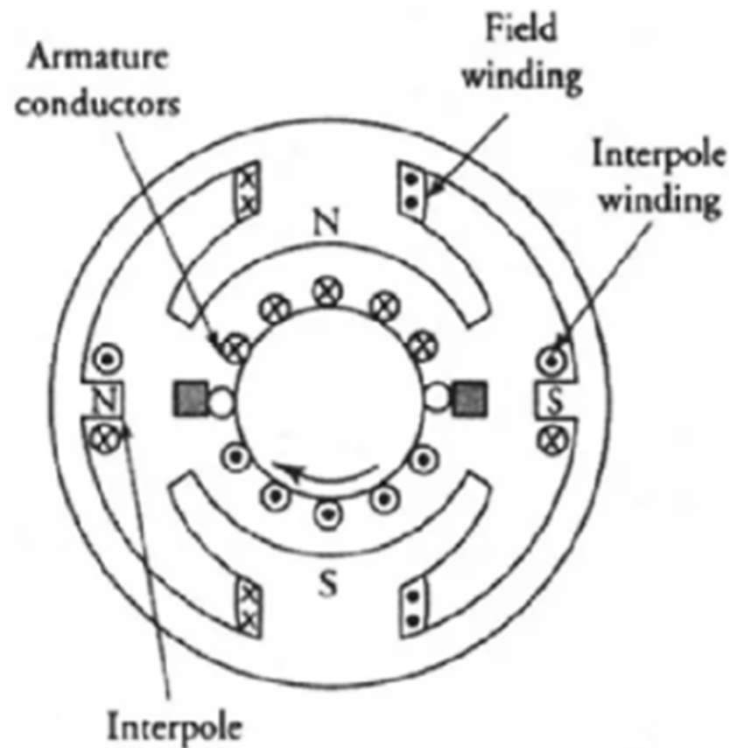
- By flattening the pole faces slightly so that the air gap is longer at the pole tips than at the centre. This increases the reluctance at the pole tip which reduces the armature cross flux
- By making trailing horn of pole piece longer than the leading horn, and cutting farther from the surface of armature so as to equalise the distribution of armature flux
- Increasing the air gap
- Providing interpoles and compensating windings

Compensating windings

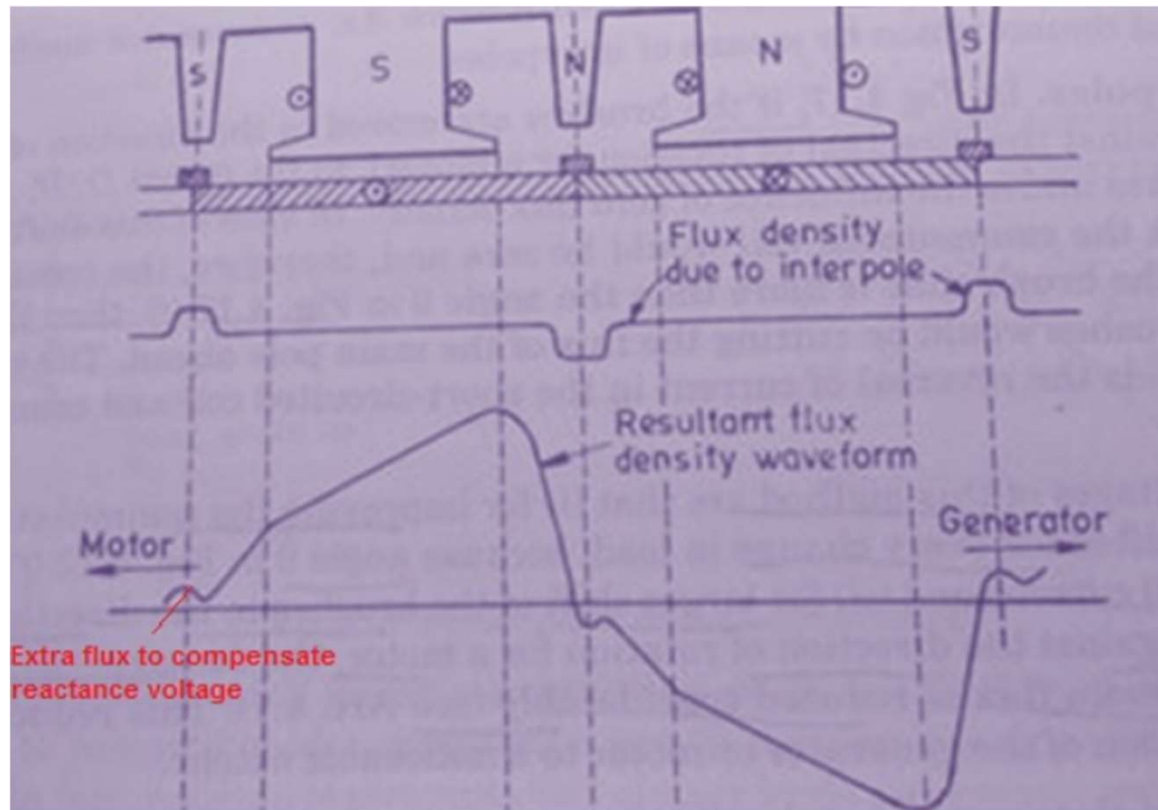


- Compensating windings are placed on the pole faces of the machine and connected in series with the armature winding in such a way that current in compensating winding conductors is equal and opposite to that in the armature winding conductors.
- Best way to reduce Armature reaction
- Expensive

Interpoles



- Additional small narrow poles provided in between the main poles in large machine to get rid of commutation problem due to armature reaction
- Interpoles are kept directly over the conductors being commutated and their polarity is same as the main pole in the direction of rotation
- Flux provided by the interpoles nullify the voltage in the coils undergoing commutation
- If the cancellation is exact there will be no sparking at the brushes.



COMMUTATION

- The Commutation process involves the change from generated alternating current to an extremely available direct current
- This reversal of current in the armature current is made possible with the help of commutator segments and brushes
- Period during which the coil remains short circuited is called commutation period T_c
- Good commutation means no sparking at brushes and the commutator surface remaining unaffected during commutation process
- Poor commutation means sparking at brushes and the commutator segment get damaged during machine operation

Poor commutation

- Can be both mechanical and electrical
- Mechanical causes – uneven commutator surface, non uniform brush pressure, vibration of brush holders etc...
- Electrical causes – increase in voltage between commutator segments, increase in current density at trailing edge of brush etc..

Commutation process

